

Giacomo Cappelletto

Boston, MA | Treviso, Italy

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PROFESSIONAL SUMMARY

Computer Engineering student at Boston University (GPA: 3.96) and Division I rower. Built full-stack products (Next.js/TypeScript, Rails), native desktop apps (Tauri/Rust), and CV/biomechanics pipelines (MMPose, MotionBERT); open to SWE, full-stack, and applied ML internships.

EDUCATION

- **Boston University, Boston, MA** 2024 – 2028
B.Sc. in Computer Engineering; GPA: 3.96
- **H-Farm International School, Venice, Italy** 2022 – 2024
International Baccalaureate (IB) Diploma *HL: Mathematics, Physics, Computer Science*

EXPERIENCE

- **Boston University College of Engineering, Boston, MA** Dec 2025 – Present
Undergraduate Researcher
 - **Biomechanics Pipeline:** Building a Python pipeline to stabilize rowing video, run single-athlete 2D keypoint inference (MMPose), lift to 3D skeletons (MotionBERT), and output per-frame joint angles/positions.
 - **Kinematic Analysis:** Computing stroke-averaged kinematics from 3D skeleton sequences to quantify coordination, range of motion, and technical consistency across strokes.
 - **Modeling:** Training a sequence-to-sequence regression model mapping time-aligned kinematics to rowing force curves, validated against instrumented on-water/erg telemetry.
- **SOCIETA CAPPELLETTO S.R.L., Treviso, Italy** May 2025 – Sep 2025
Full-Stack Developer (Tauri/React/Rust Inventory System Rebuild)
 - **Migration & Performance:** Rebuilt an Electron inventory app using Tauri (Rust) + React, reducing app size by 70%. Parallelized API calls, improving search by 43%.
 - **Advanced Features:** Built audit history (Firestore), real-time multi-location inventory reconciliation, and GraphQL-optimized Shopify queries; added SKU instant search with fallback paths.
 - **Operational Impact:** Reduced memory footprint by 80% via a native Rust backend; improved reliability and eliminated search delays with immediate inventory updates.
- **TickIT GmbH, Boston, MA (Remote)** Oct 2024 – Jan 2025
Full-Stack Engineer (Ticketing Platform Startup)
 - **Overview:** Built responsive UIs (Next.js, shadcn/ui, Tailwind) and backend features (Rails, PostgreSQL); shipped scalable API integrations for a ticketing platform.
 - **Contribution:** Contributed architecture for ticket authentication and improved Agile workflows while ramping quickly on a new stack.
- **SOCIETA CAPPELLETTO S.R.L., Treviso, Italy** May 2024 – Jul 2024
Full-Stack Developer (Inventory Management System)
 - **Project:** Built an Electron + React + Node.js app integrating Shopify API + Firestore to sync online inventory across two locations.
 - **Impact:** Eliminated inventory discrepancies and cut checkout manual work by 80%.
- **H-Farm International School, Treviso, Italy** Mar 2024 – Apr 2024
Python Developer (Meeting Scheduling App)
 - **Application:** Built a Python/Streamlit app that reads school database timetables and auto-drafts meeting-request emails using local LLMs (Ollama).
 - **Collaboration:** Partnered with faculty to tailor workflows and integrate with existing processes, improving scheduling efficiency.

- **DLF Treviso, Treviso, Italy**

Sep 2023 – Nov 2023

iOS Developer (Rowing Logbook App)

- **Development:** Built a SwiftUI iOS training log with Swift Charts for session tracking and performance analytics.
- **Process:** Owned end-to-end development and iterated from athlete feedback.

- **E-TRASH, Treviso, Italy**

Feb 2023 – Jun 2023

CAD Engineer Intern

- **CAD Modeling:** Designed the first prototype of an intelligent trash bin in Autodesk Fusion 360, focusing on structural integrity.
- **Optimization:** Applied topology optimization and load testing; produced Blender renders for presentations and iteration.

PROJECTS

- **NoteWorthy – Automated Note Conversion Platform:** Built noteworthyy.xyz, a Next.js/TypeScript app that converts handwritten notes into styled PDFs and LaTeX. Integrated Google AI Studio, GitHub/Google OAuth, and a Dockerized LaTeX compiler; deployed via Cloud Run and Vercel.
- **CIFAR-10 Image Recognition with Neural Architecture Search:** Built a TensorFlow/Keras CIFAR-10 classifier and neural architecture search pipeline to generate and evaluate CNN candidates. Automated training, metric logging, and best-model selection for extended training.

SKILLS

- **Languages:** TypeScript/JavaScript, Python, Rust, Ruby, C/C++, Swift, Java
- Frameworks & Libraries:** Next.js, React, Ruby on Rails, Node.js, Tauri, SwiftUI, TensorFlow/Keras
- Tools & Platforms:** Git/GitHub, Tailwind CSS, PostgreSQL, Docker, Firebase, Shopify API (GraphQL), Streamlit, Ollama, Google Cloud Run, Vercel